Instructor Example

An airline makes 200 reservations for a flight that holds 185 passengers. The probability that a passenger arrives for the flight is 0.9, and the passengers are assumed to be independent.

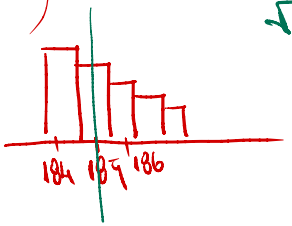
- Approximate the probability that all the passengers who arrive can be seated.
- Approximate the probability that the flight has empty seats.

X = # of passengers modeled by Binomial Dist. (0.9)

$$* \quad P(X \leq 185) = \sum_{0}^{185} \binom{200}{i} (0.9^{i})(0.1^{200-i})$$

$$Z \rightarrow \frac{185+0.5-200\times 0.9}{\sqrt{200\times 0.9\times 0.1}} = 1.3 \rightarrow P(Z \leq 1.3) = 0.9032$$

$$P(X \leq 185) \longrightarrow Z = \frac{X - np}{\sqrt{np(1-p)}}$$



$$* \quad P(X < 185) \longrightarrow Z = \frac{185-0.5-200\times 0.9}{\sqrt{200\times 0.9\times 0.1}} = 1.06 = P(Z \leq 1.06) = 0.8554$$

Definition: Lognormal R.V.

- Variables in a system sometimes follow an exponential relationship as x = exp(w). If W has a normal distribution. In that case, the distribution of X is called a lognormal distribution.

w is a normal distribution with mean θ and variance ω^2

$X = \exp(w) \rightarrow F_X(x) = P(X \leq x) = P(\exp(w) \leq x) = P(w \leq \ln(x))$

$P\left(Z \leq \frac{\ln(x) - \theta}{\omega}\right) = \Phi\left(\frac{\ln(x) - \theta}{\omega}\right)$

$\rightarrow$ CDF of standard normal distribution

Lognormal Distribution

Let W have a normal distribution with mean θ and variance ω^2; then $X = \exp(W)$ is a **lognormal random variable** with probability density function

$$f(x) = \frac{1}{x\omega\sqrt{2\pi}} \exp\left[-\frac{(\ln(x)-\theta)^2}{2\omega^2}\right] \quad 0 < x < \infty$$

The mean and variance of X are

$$E(X) = e^{\theta+\omega^2/2} \quad \text{and} \quad V(X) = e^{2\theta+\omega^2}\left(e^{\omega^2} - 1\right)$$

Instructor Example

The lifetime (in hours) of a semiconductor laser has a lognormal distribution with θ = 10 and ω = 1.5. What is the probability that the lifetime exceeds 10,000 hours?

$$P(X>10000) = 1-P(X\leq 10000) \Rightarrow 1-P(\exp(W)\leq 10000)$$
$$= 1-P(W\leq \ln(10000))$$
$$= 1-\Phi\left(\frac{\ln(10000)-10}{1.5}\right)$$
$$1-\Phi(-0.92) = 1-(1-\Phi(0.92))$$
$$= \Phi(0.92) = \underline{0.6985}$$

What lifetime is exceeded by 99% of lasers?

$$P(X>x) = 0.99 \Rightarrow 1-\Phi\left(\frac{\ln(x)-10}{1.5}\right) = 0.99 \rightarrow z = -2.33$$

$$\frac{\ln(x)-10}{1.5} = -2.33 \Rightarrow x = \exp(6.505) = \underline{668.48 \text{ hours}}$$

Instructor Example (why lognormal?)

Suppose that the length of stay (in hours) at a hospital emergency department is modeled with a lognormal random variable X with θ = 1.5 and ω=0.4

Mean and Variance

$$\mu = E(X) = e^{\theta+\omega^2/2} = e^{1.5+0.16/2} = e^{1.58} = 4.8550$$

$$V(X) = e^{2\theta+\omega^2}(e^{\omega^2}-1) = e^{3+0.16}(e^{0.16}-1) = 4.0898$$

Comment on the difference between the probability P(X < 0) calculated from this lognormal distribution and a normal distribution with the same mean and variance.

$$P(X<0) \xrightarrow{\text{modeled by lognormal}} P(X<0) = 0$$

$$P(X<0) \xrightarrow{\text{modeled by normal}} P\left(Z < \frac{0-4.855}{\sqrt{4.0898}}\right)$$

$$P(Z<-2.4) = 1-0.9918$$

$$= 0.0082$$

Special Topic: Mixed R.V.

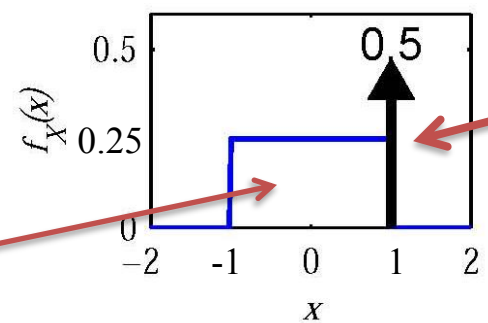
- X, Daily rainfall, measured in cm
- Y, The delay time in a queue at an ice cream shop

A mixed random variable consists of

- At least one sample value with non-zero probability (discrete)
- A sample value that cover an interval (continuous)

We can use the tools from previous chapters to analyze them

Discrete value



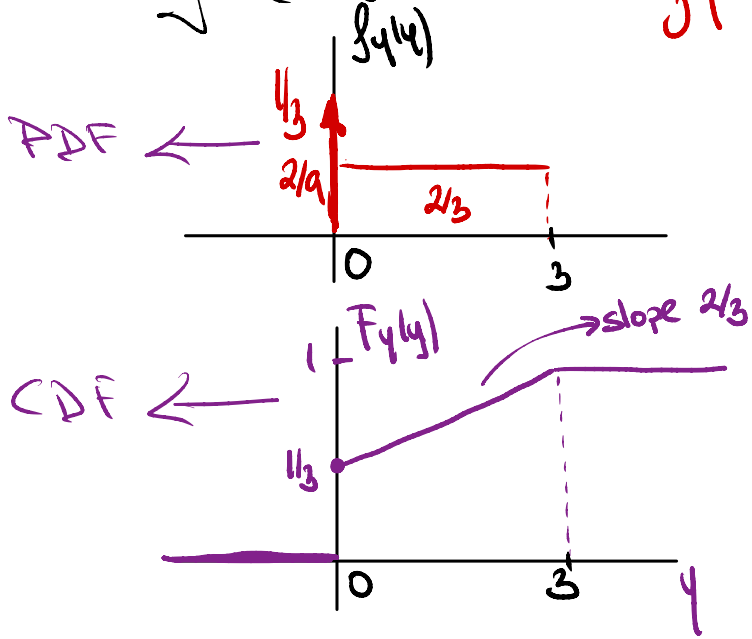
Continuous value

Sample PDF of a Mixed R.V.

Example-3.12

Observe someone dialing a telephone and record the duration of the call. In a simple model of the experiment, 1/3 of the calls never begin either because no one answers or the line is busy. The duration of these calls is 0 minutes. Otherwise, with probability 2/3, a call duration is uniformly distributed between 0 and 3 minutes. Let Y denote the call duration. Find the CDF $F_Y(y)$, the PDF $f_Y(y)$, and the expected value E[Y].

y = call duration



$$f_Y(y) = \begin{cases} 1/3 & y=0 \\ 2/3 & 0<y\leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = 0 \times 1/3 + \int_0^3 x \cdot 2/9 \, dx$$